Project Sharknado: Evaluating the Risk of Coastal-Origin Tornadoes in the United States

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Executive Summary

Figure 1: This is a fictional mockup of what a sharknado could look like if they were actually real.

Purpose of Report

The insurance agency has seen a rise in property payouts driven by the intensifying effects of climate change on extreme weather. In the wake of the 'Sharknado' phenomenon sparked by the 2013 film, the agency is now investigating whether the increasing frequency of severe weather could lead to a higher likelihood of shark-infested weather events (as depicted in Figure 1). The purpose of this report is to

- **Determine the likelihood of sharknadoes forming.**
- **Determine the economic impact of sharknadoes.**

Background

Historically, the majority of tornadoes have occurred in the Central United States, a region nicknamed "Tornado Alley". Since this region is not near any coasts, the plausibility of a tornado picking up a shark out of coastal waters is low. However, there have been a few recorded tornadoes near the coast, leading the agency to wonder if sharknadoes are not an impossibility.

While there has not been any formal scientific literature that has studied the likelihood and impacts of tornadoes carrying sharks, one scholar, Dr. Kim Martini, entertained the idea and provided an analysis on the likelihood of a sharknado with her knowledge in marine biology and physics. Through logical reasoning, Martini concluded that while a typical tornado travels over enough geographic area and can be strong enough to keep a shark airborne, it is highly unlikely that a tornado could lift the shark out of the water in the first place.

For the sake of still conducting a statistical analysis, this report ignores this physical impossibility.

Methods

75 years of tornado storm data (1950-2025) from the National Weather Service was analyzed to assess the likelihood and economic impact of a sharknado. After cleaning the data, a total of 31,345 storms were classified by path trajectory and shark-lifting potential (based on the Fujita scale) and analyzed for sharknado-likeness and total property damage.

A tornado's *path trajectory* was classified as one of three labels (see Figure A.2):

A tornado's *shark-lifting potential* was classified as one of two labels:

- **Low Potential**: tornado was given an F1 or F2 rating on the (Enhanced) Fujita Scale
- **High Potential**: tornado was given an F3, F4, or F5 rating on the (Enhanced) Fujita Scale

Findings & Conclusions

Throughout the analysis, storms classified as "Moving Inland" and "High [shark-lifting] Potential" were of the highest interest as they exhibit characteristics most similar to a sharknado. Of the 31,345 recorded storms, only 2 were classified as both "Moving Inland" and "High Potential" for lifting sharks. While the Atlantic and Gulf coasts were the primary region of origin for "Moving Inland" storms, these events statistically lacked the same shark-lifting potential of storms originating inland. In terms of economic impacts, "Moving Inland" storm with high

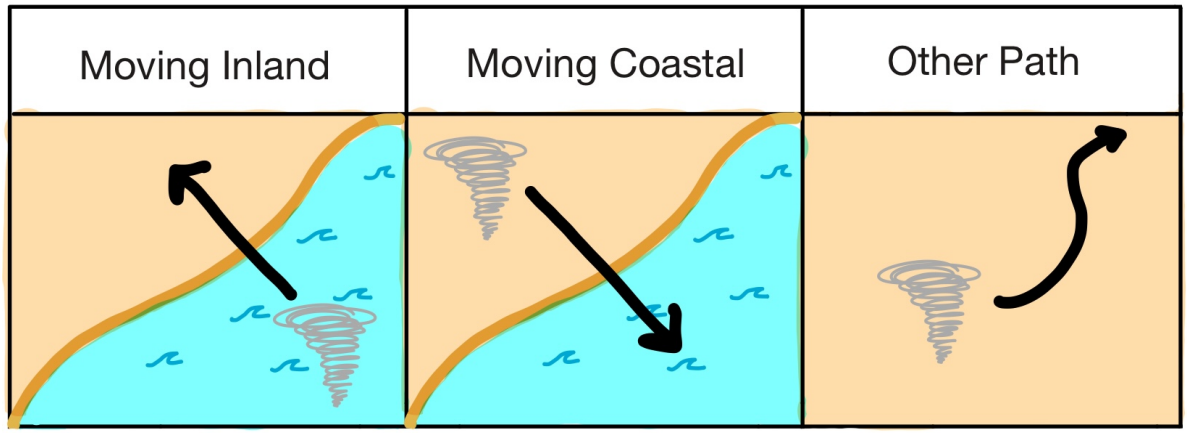

Figure 2: This is a diagram illustrating the movement of tornadoes relative to the coast and inland, with path trajectory labels of "Moving Inland", "Moving Coastal", and "Other Path".

shark-lifting potential account for less that 0.01% of total historical property damage ($183K over 75 years).

Reccome

- **Treat Marine-Origin Events as Rare Events:** High-intensity "Moving Inland" tornadoes remain an extreme statistical rarity, with only two recorded occurrences in 75 years. These should be categorized as low-probability, "long-tail" risks rather than primary drivers of annual catastrophe loss.
- **Maintain Standard Inland Risk Models**: With 97.86% of historical property damage occurring in "Other Path" trajectories, the agency should continue prioritizing traditional inland tornado risk models over specialized "Sharknado" or marine-origin coverages.
- **Monitor Southeast Regional Shifts:** While national "Sharknado" risk is negligible, the Gulf and Atlantic coasts account for the vast majority of coastal-to-inland storm paths. The agency should periodically track storm trends in the Southeastern U.S., where climate change *may* be increasing the frequency of atypical, high-intensity coastal events.

1 Introduction

Before the release of the movie, Sharknado (Ferrante 2013), most people probably had not thought about the possibility of a tornado close enough to water and so powerful that it could pick up a shark and hurl it onto land. However, with the recent rise in extreme weather events due to climate change ("Tornadoes and Climate Change. National Geographic Education" 2023), Anthony Ferrante's idea of a sharknado in today's climate environment may not be as laughable as it was back in 2013. Thus, there exists a motivation to investigate if sharknadoes could become a real threat to the safety of our communities (and the profitability of insurance companies).

Historically, we have seen tornadoes primarily occurring in the Central United States region, which has gained the nickname "Tornado Alley" (see Figure 1.1). Tornadoes typically last for a few minutes and travel for short distances of 1 to 5 miles, though long-track events can exceed 50 miles. The intensity of a tornado is ranked on a scale of EF0 to EF5; while the original Fujita (F) scale was used historically, the modern Enhanced Fujita (EF) scale focuses on degrees of damage to better estimate wind speeds (National Weather Service 2025). Under this system, an EF1 tornado is strong enough to do moderate damage, such as stripping roof surfaces or upsetting mobile homes, while an EF5 tornado is strong enough to lift incredibly heavy objects—including automobiles and entire frame houses—and cause incredible structural devastation (see Figure 1.2).

Additionally, the 75 years of tornado data provided by the NOAA (DOC/NOAA/NESDIS/NCEI 2026) (which we will later analyze in more detail) records a small, but not impossible, set of coastal tornado occurrences. With various species of sharks existing along the Atlantic, Pacific, and Gulf Coasts of the US, a coastal tornado's "inventory" of sharks is not limited to any specific region (Fisheries 2025). Furthermore, with an increasing number of tornadoes in atypical regions such as the Southeastern United States ("Tornadoes and Climate Change. National Geographic Education" 2023), tornadoes' proximity to sharks and ability to pick them up may be increasing as well.

While there has not been any formal scientific literature that has studied the likelihood and impacts of tornadoes carrying sharks, one scholar, Dr. Kim Martini, entertained the idea and provided an analysis on the likelihood of a sharknado with her knowledge in marine biology and physics (Martini 2013). Her analysis resulted in several conclusions:

1. **A typical tornado flying over a typical reef could pick up about 150 sharks**.
   Rationale: A typical tornado is about 0.1 miles wide, travels about 5 miles, and covers an

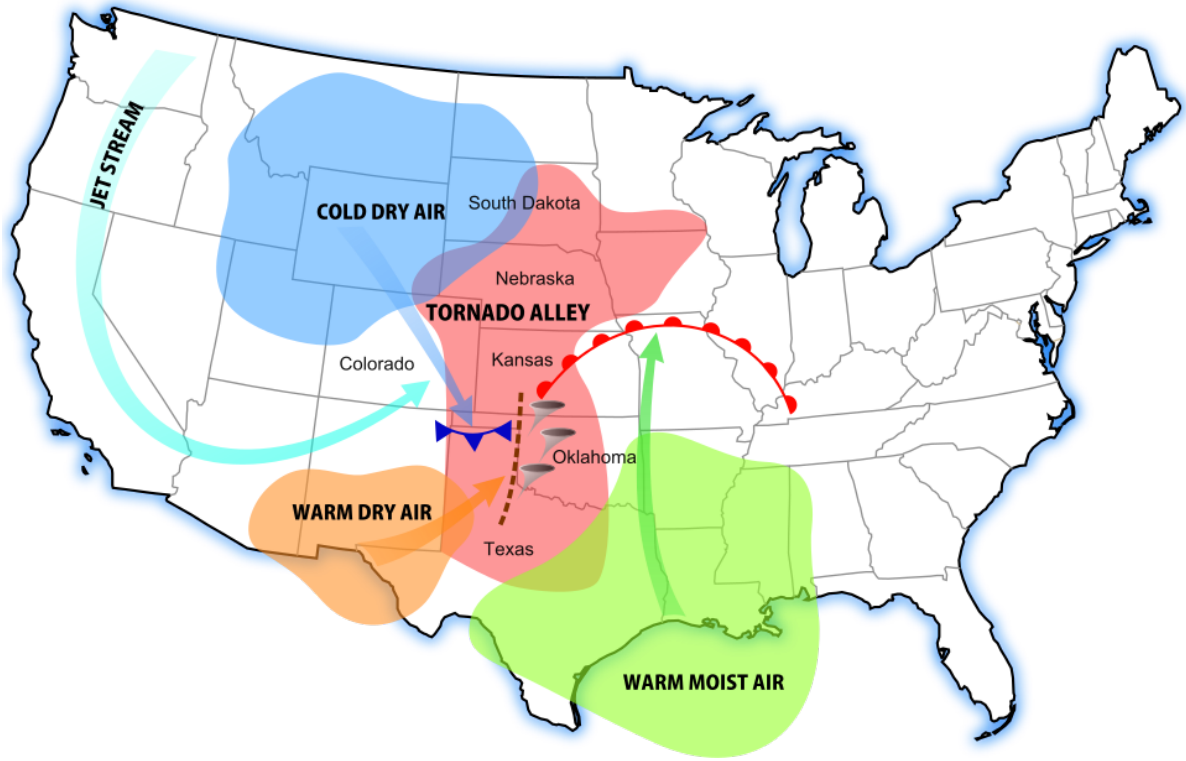

Figure 1.1: This is map highlights the Central United States, historically referred to as Tornado Alley ("Tornado Alley" 2026).

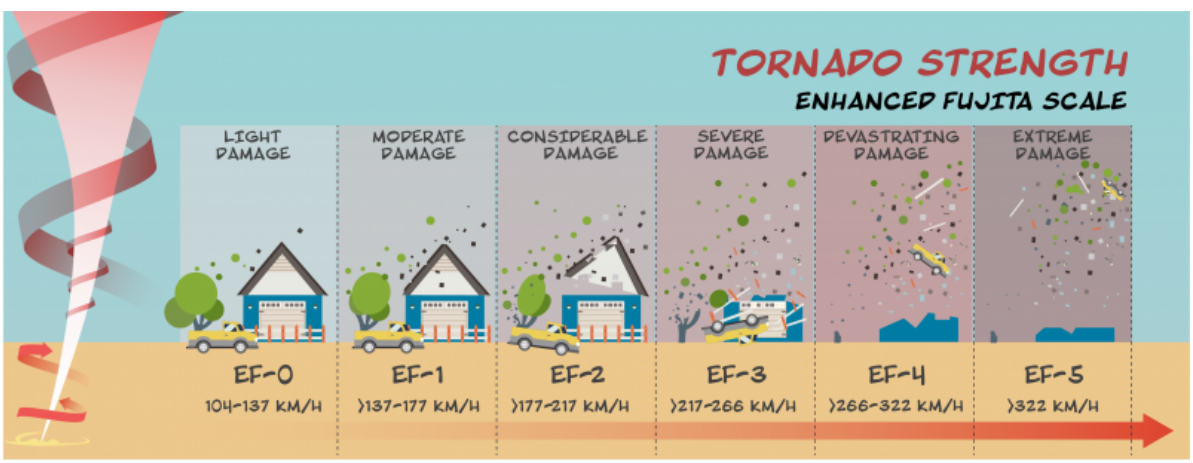

Figure 1.2: This is a diagram illustrating the strength, wind speed, and damage typically caused by a tornado at each classification on the Enhanced Fujita Scale ("What Is a Tornado? Let's Talk Science" 2020).

area of 0.5 square miles. A healthy reef typically has a shark density of about 300 sharks per square mile. However, she notes that one of the largest tornadoes in history was 2.6 miles wide and traveled 52 miles ("2004 Hallam Tornado" 2026), making it hypothetically capable of picking up over 40,000 sharks.

2. **It is highly unlikely that a tornado could suck a shark out of the water.**
   Rationale: Tornadic water spouts (see Figure 1.3) don't typically suck up water, and tagged sharks typically dive deep when there are approaching thunderstorms (which can cause tornadoes).
3. **Intense tornadoes are strong enough to keep a shark airborne.**
   Rationale: In the event that a tornado does pick up a shark (e.g., a shark hypothetically jumps out of the water as a tornado passes over contrary to typical observed behavior), the tornado would need to have strong enough vertical wind speeds to combat the force of gravity on the shark. With basic physics equations and conservative estimates, the minimum upward wind speed required to keep an average Great White Shark in the air is 127 mph, which a strong EF4 tornado has shown to exhibit. Additionally, smaller sharks are even lighter and would require less force to keep them airborne.

Figure 1.3: This is a photo of a tornadic water spout, essentially a tornado that forms over water or moves from land to water.

Using this logic, we can assume that EF3 tornadoes (and larger) are strong enough to pick up small sharks, given that the shark was transported from the water into the tornado by some odd means. This transportation of the shark from water to tornado already makes the occurrence of a sharknado highly unlikely. However, for the purposes of this report, we will ignore this improbability and assume the physical transport of sharks is possible, so we can continue analyzing the geographic and economic frequency of the storm paths themselves.

While tornado intensity is well-studied and the ability for a high intensity tornado to pick up a shark has been logically proven, there is a gap in the analysis of the two together. To ensure the profitability of the insurance agency, we need to know how serious the threat of sharknadoes is to the insurance agency's bottom line given the warming climate and shifting trends in tornado activity. Thus, we will use the historical data found in the NOAA Storm Events Database to evaluate the likelihood and economic impact of a high-intensity tornado with a coastal origin moving inland (since these are the types of tornadoes that have the ability to transport sharks from the ocean) to answer the research question:

What is the historical frequency, probability, and economic impact of high-intensity tornadoes (F3, F4, and F5) originating in coastal areas (Atlantic, Pacific, and Gulf of Mexico) and moving inland, compared to those following other trajectories?

2 Methodology

In order to answer the research question (stated at the end of Chapter 1), we must

1. Define what qualifies as a storm as a potential "sharknado"
2. Use publicly available historical data to determine the frequency, probability, and economic impact of sharknadoes

2.1 Defining Sharknado

For a tornado to be a potential "sharknado", it has to have a few characteristics:

1. Originates on the coast
2. Strong enough to sustain lifting a shark
3. Path trajectory moves to the inland

From this, we have to also answer "What is the 'coast' vs what is 'inland'?" and "What indicates a tornado is 'strong enough' to sustain lifting a shark?"

What is the "coast" vs what is "inland"?

For the boundary between the coast and open water, the general definition of a state's coastal zone extends 3 nautical miles into the ocean from the shore (with some states having their own exceptions). For the boundary between the coast and the inland, states define their coastal zone as an extension anywhere from 1,000 feet to 5 miles inland of the shoreline (Coastal Management n.d.).

Since there is no singular standard, a 3-mile buffer was used to define the coastal zone along the coastline. This buffer is equally centered over the coastline, resulting in a zone that extends 1.5 miles into the ocean and 1.5 miles onto the land. Any point within this buffer zone is considered the "Coast" for the purposes of this analysis. Any point further inland than the buffer zone is considered "Inland". Section A.1.1 details how this coastal buffer zone was represented in the data.

What indicates a tornado is "strong enough" to sustain lifting a shark?

Based on the physical analysis done by Dr. Kim Martini (discussed in Chapter 1), a strong EF4 tornado has enough vertical wind speed to keep a great white shark suspended in the air. Because great white sharks are significantly larger than many other shark species, it's logical

to believe an EF3 tornado has enough vertical wind speed to suspend much smaller sharks in the air. Since a shark is a shark, no matter how big or small, this analysis defines tornadoes classified as (E)F3 and higher on the (Enhanced) Fujita Scale as "strong enough" to sustain lifting a shark. Section A.1.2.3 details how shark-lifting potential is identified in the data.

> **Important**
>
> For the purposes of this analysis, tornado classifications from the Fujita and Enhanced Fujita Scales were treated synonymously (i.e., EF5 = F5). Section 4.4 discusses potential limitations of this.

2.2 Data Sources

The primary data for this business report comes from the National Oceanic and Atmospheric (NOAA) Storm Events Database, which includes 75 years (1950 to 2025) of records of various types of weather events. This dataset was selected as the primary source because it is one of the most comprehensive official records of tornadic activity in the United States, as it provides standardized metrics for storm intensity, location, and property damage.

Additionally, a shapefile of the 2025 United States coastline was downloaded into R using the R package `tigris`. In the analysis, this shapefile is used to establish the study's geographic boundaries between coast and the inland.

For more documentation on these data sources, please refer to Appendix B.

2.3 Data Pipeline

> **Note**
>
> - See Figure A.3 for a diagram of the full data pipeline.
> - See Appendix A for more technical details on this methodology.

2.3.1 US Coastline Data

To implement the above definition of "Coast" and "Inland" (see Section 2.1), the R package `tigris` was used to provide a shapefile of the US Coastline. Once the original coastline shapefile was imported, it was filtered to only include the Atlantic, Pacific, and Gulf of Mexico coasts as requested in the original scenario prompt. The units of the coastline data were converted from degrees to meters. A 3-mile wide boundary was equally centered over the coastline, resulting in

a zone that extends 1.5 miles into the ocean and 1.5 miles onto the land. Any point within this buffer zone was considered the "Coast" for the purposes of this analysis. Any point further inland than the buffer zone was considered "Inland". A visual representation of this data can be seen in Figure A.1.

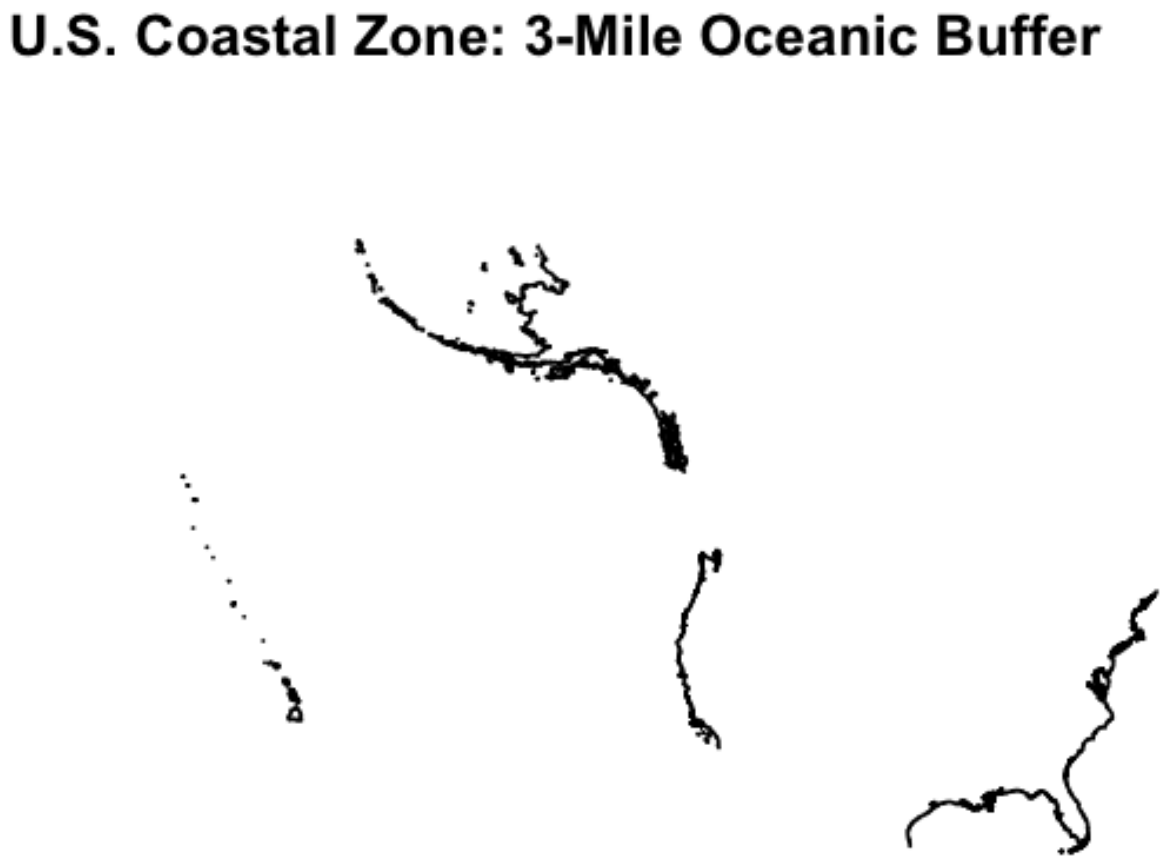

Figure 2.1: This is a visual representation of the US Coastline 3-mile buffer zone generated in the code.

2.3.2 NOAA Storm Event Database

2.3.2.1 Pre-cleaning & Filtering

To begin processing the NOAA Storm Events Database, the data was filtered to only include tornado weather events. Additionally, any entries that were missing necessary data were removed.

2.3.2.2 Geographical Classifications

Beginning the work needed to classify tornado path trajectories, each tornado segment's starting/ending coasts were identified as "Atlantic", "Pacific", "Gulf", or N/A. Each tornado segment's starting/ending location was also identified as "Coastal" or "Inland".

2.3.2.3 Aggregate Storm Segments

NOAA Storm Events Database is structured so that a single storm may be recorded as multiple different storm segments. A new storm segment is created when the tornado enters a new county or zone, resulting in multiple data rows corresponding to one storm. Thus, segments from the same storm were identified and stitched back together into a single row of data.

From here, each tornado's path trajectory was classified as "Moving Inland", "Moving Coastal", or "Other Path" based on the tornado's start/end locations. Below further describes the characteristics of each path trajectory label, along with an illustration in Figure A.2.

- **Moving Inland**: tornado originated on the coast and moved inland.
- **Moving Coastal**: tornado originated inland and moved to the coast.
- **Other Path**: tornado originated either inland or on the coast and stayed in the same region of origin.

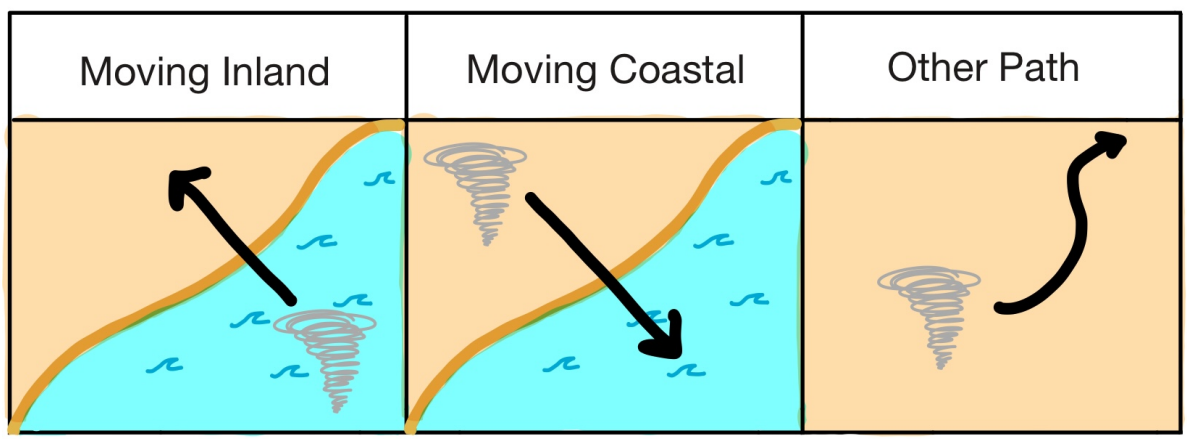

Figure 2.2: This is a diagram illustrating the movement of tornadoes relative to the coast and inland, with path trajectory labels of "Moving Inland", "Moving Coastal", and "Other Path".

Finally, each tornado's shark-lifting potential was classified as "High Potential" or "Low Potential", based on the storm's (Enhanced) Fujita Scale RANKING. Below further describes the characteristics of these labels:

- **Low Potential**: tornado was given an F1 or F2 rating on the (Enhanced) Fujita Scale
- **High Potential**: tornado was given an F3, F4, or F5 rating on the (Enhanced) Fujita Scale

2.4 Statistical Analysis Approach

The statistical analysis provided in the next section uses an empirical probability model and inferential statistics to convert raw storm counts into actionable risk metrics. The study calculates the likelihood of a "sharknado" by determining the ratio of high-intensity, "Moving Inland" tornadoes against the total study population. This approach is justified as it directly answers the research question regarding historical probability and provides a standardized frequency for insurance risk modeling. To address the uncertainty in small sample sizes, a 95% confidence interval was calculated for sharknado frequency. This moves the analysis beyond simple point estimation to provide a statistical range of certainty, which is critical for actuarial worst-case planning.

While more advanced predictive modeling (such as logistic regression) was initially considered, the extreme scarcity of the target event ($n = 2$) made more complex methods statistically inappropriate. A thorough analysis of descriptive statistics along with confidence intervals provided a straightforward evaluation method of the historical frequency of sharknado-like events.

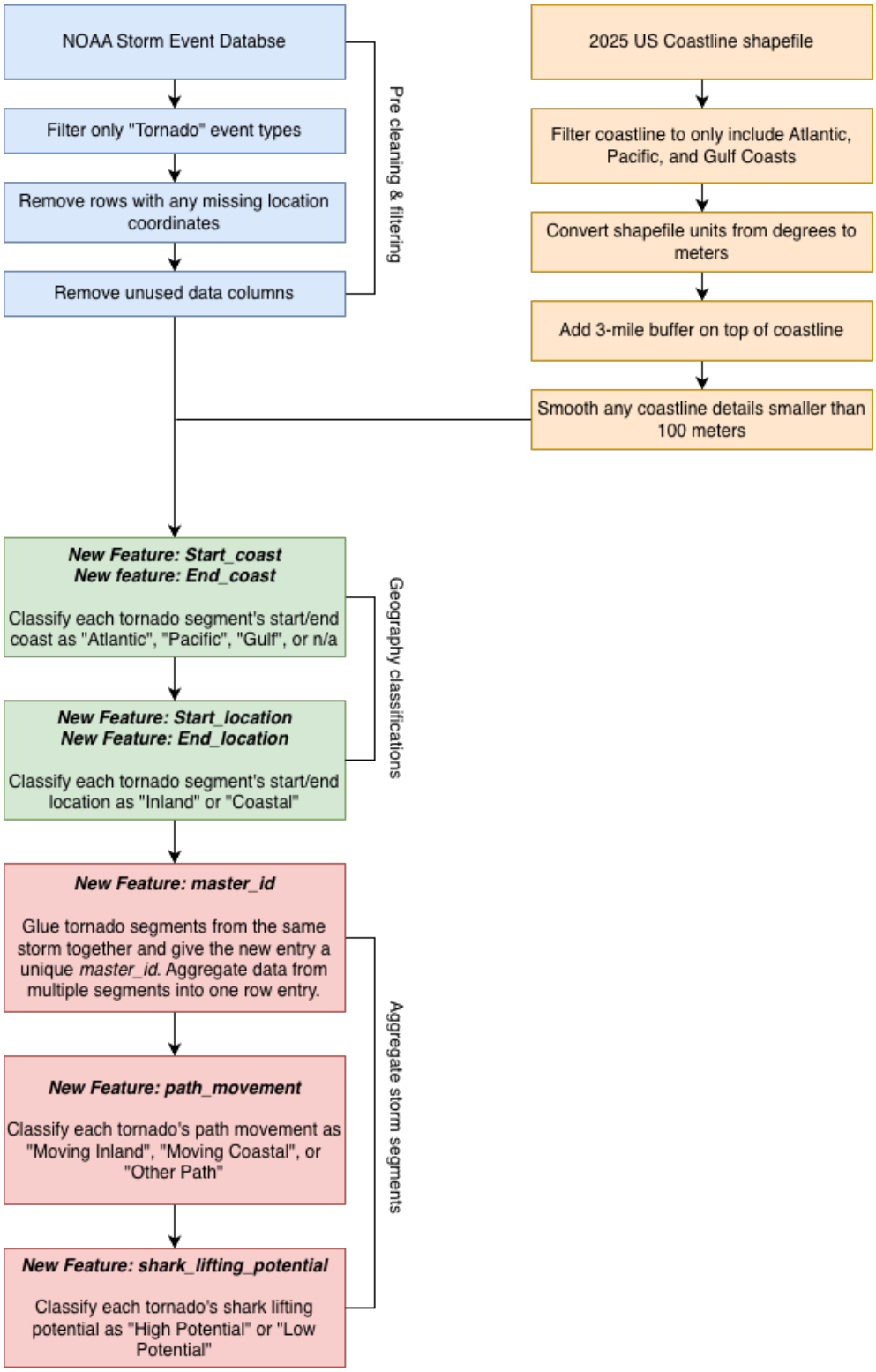

Figure 2.3: This is a diagram illustrating the steps used to process data and engineer features from the NOAA Storm Events Database and US Coastline shapefile.

3 Results

3.1 Overview of the Study Population

The analysis of the tornado data from the NOAA Storm Events Database spans from 1950 through 2025. After combining individual storm segments into complete tornado events, a total of 31,345 unique tornado events were identified across the United States. Of these events, 761 started within the defined 3-mile coastal zone, and only 73 of those ultimately moved from the coast to the inland. These events were categorized by their trajectory to determine the potential for transporting sharks. The population breakdown is as follows in Table 3.1:

Initial findings indicate that "High Potential" and "Moving Inland" events are a minority of total tornado occurrences, with only **2 total** tornadoes meeting the criteria for "High Potential" and "Moving Inland". While these unique events are quite rare, they are certainly not impossible and present a specific kind of threat that current risk models might be missing.

Furthermore, the analysis identified an empirical frequency of 0.0064% for high-intensity coastal-to-inland trajectories. To determine the precision of this estimate, a 95% confidence interval was calculated. The resulting interval, ranging from 0.0017% to 0.023%, indicates that while the true probability is extremely low, the upper bound remains nearly four times higher than the point estimate.

3.2 Path Movement & Intensity Analysis

To further evaluate the relationship between tornado path trajectory and intensity, we can visualize the metrics from the previous Table 3.1. In Figure 3.1 below, we can see a noticeable difference in tornadoes that remained completely inland ("Other Path") compared to tornadoes that either originated or ended on the coasts. While "Moving Inland" trajectories are the primary focus when considering the risk of sharknadoes, only **2.7%** of these tornadoes moving from the coast to the inland reached high-potential intensity. In contrast, "Other Path" tornadoes exhibited a much larger high-potential rate of **12.2%**.

This suggests that while the "Moving Inland" path is geographically unique, the prime atmospheric conditions needed for F3/F4/F5 tornadoes are far more prevalent once the storm has moved more inland over land. For insurance risk modeling, this indicates that while

Table 3.1

Tornado Path Trajectory and Shark-Lifting Potential
1950-2025

Shark-Lifting Potential	Event Count	Proportion of Trajectory
Moving Coastal		
High Potential	2	3.2%
Low Potential	60	96.8%
Moving Inland		
High Potential	2	2.7%
Low Potential	71	97.3%
Other Path		
High Potential	3,799	12.2%
Low Potential	27,411	87.8%

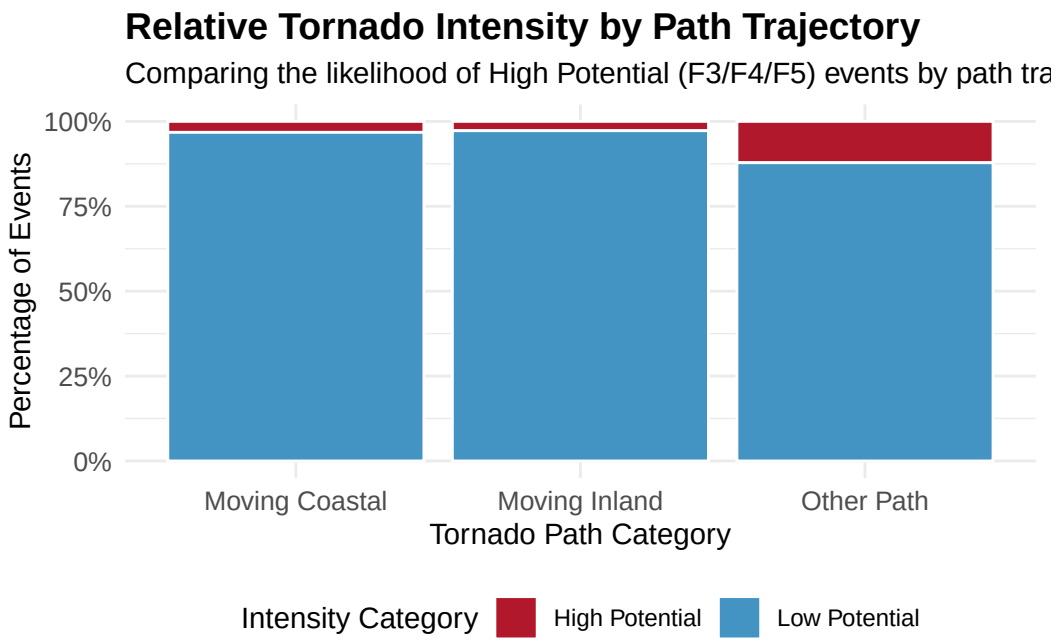

Figure 3.1: Categorized all tornadoes by path trajectory and visualized the percentages of high vs. low shark-lifting potential within each path trajectory group.

coastal-origin tornadoes are *capable* of transporting sharks, they are statistically much less likely to reach the extreme wind speeds of their inland counterparts.

3.3 Regional Risk Distribution

To identify likely areas where we could find sharknadoes forming, the data was segmented by coastal region (Atlantic, Gulf, and Pacific). The analysis in Figure 3.2 reveals a geographic imbalance in risk.

The **Gulf Coast** and **Atlantic Coast** account for the vast majority of "Moving Inland" events, with each coast responsible for 36 "Moving Inland" tornadoes. In contrast, the **Pacific Coast** only recorded **1 total** "Moving Inland" tornado, which was identified as "Low Potential" intensity. The **Atlantic Coast** was the only coastal region that had recorded a "Moving Inland" tornado with **high shark-lifting potential** (2 total occurrences recorded). Overall, though, the majority of the "Moving Inland" tornadoes produced by any of the 3 coasts were identified as "Low Potential" intensity.

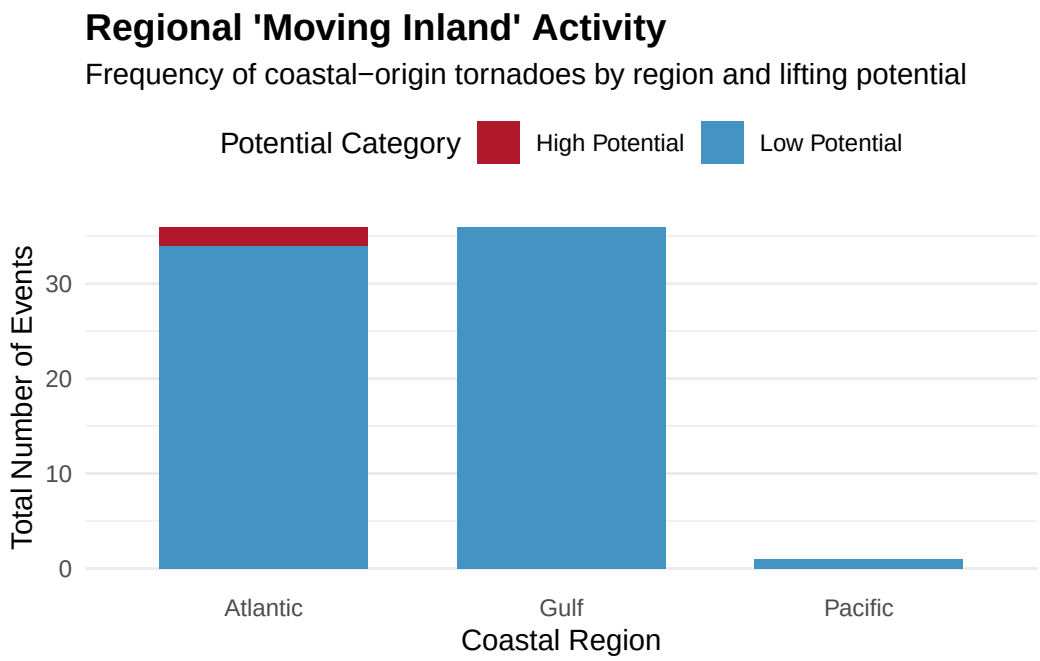

Figure 3.2: Categorized only 'Moving Inland' trajectory tornadoes by coastal region and visualized the percentages of high vs. low shark-lifting potential within each coastal region group.

This confirms that sharknado potential is a regional phenomenon, rather than a national one, with the Atlantic and Gulf Coasts requiring significantly different actuarial considerations than the West Coast.

Table 3.2

Economic Impact Analysis by Tornado Trajectory
Property damage and frequency (1950-2025)

Shark-Lifting Potential	Total Events	Cumulative Damage	Avg. Damage per Event
Moving Coastal			
Low Potential	60	$41.79M	$696.53K
High Potential	2	$9.47M	$4.73M
Moving Inland			
Low Potential	71	$24.79M	$349.21K
High Potential	2	$183.00K	$91.50K
Other Path			
Low Potential	27,411	$2.58B	$93.97K
High Potential	3,799	$682.77M	$179.72K

3.4 Economic Impact

The final phase of this analysis evaluated the economic impact of tornadoes based on their path trajectory. Across the 75-year study period, the total property damage recorded for all events summed to **$3.33B** with an average of $106,389 per tornado. The summary of economic impact by tornado trajectory and shark-lifting potential is in Table 3.2 below:

As visualized in Figure 3.3 below, tornadoes following "Other Path" trajectories have contributed to **97.86%** of property damage in the last 75 years, which makes sense since they are the most commonly occurring tornadoes. In contrast, tornadoes that originated on the coast and moved inland ("Moving Inland") have contributed to only **0.75%** of all recorded tornado property damage, with only **$183K** in damages resulting from high shark-lifting potential tornadoes under the "Moving Inland" trajectory category.

Based on the visual analysis provided in Figure 3.4, a hasty conclusion would be that tornadoes originating inland and moving to the coast ("Moving Coastal") result in the most property damage on average. Based on the calculations, this is technically true. However, it's important to note that for "Moving Coastal" and "Moving Inland" tornadoes classified as high shark-lifting potential, the population consists only of **2 total recorded events** for both groups.

Ultimately, when looking at the tornadoes with sharknado characteristics ("Moving Inland" with high shark-lifting potential), only **$183K** in damages has been inflicted in the last 75 years with an average damage of **$91.5K per storm**. From a risk management perspective, this indicates that while high-intensity coastal-to-inland events are physically possible, their

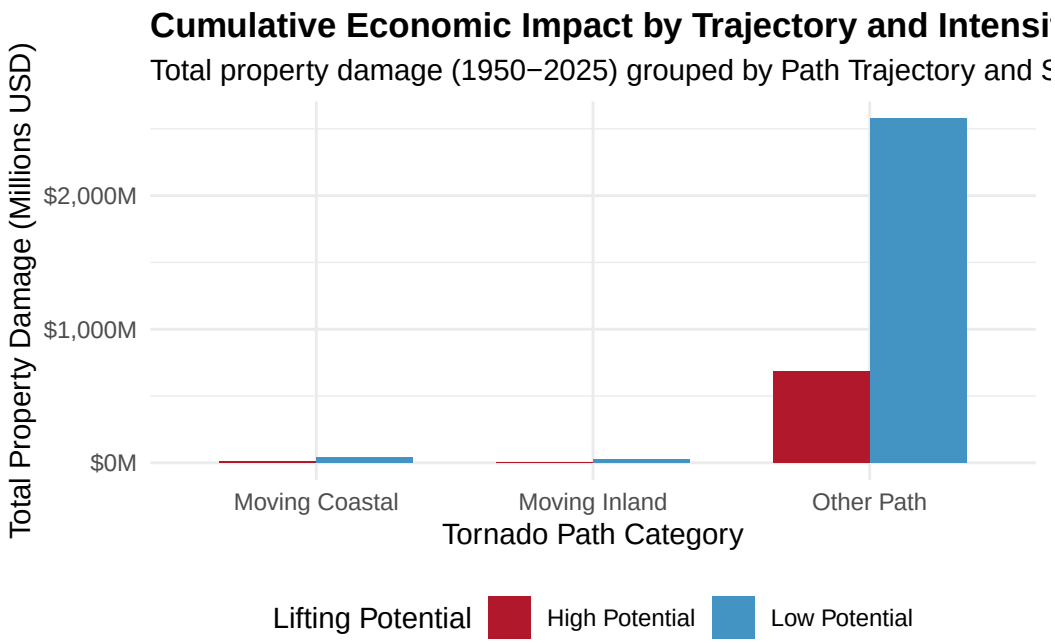

Figure 3.3: Categorized all tornadoes by trajectory and visualized the total property damage inflicted by high vs. low shark-lifting potential tornadoes within each path trajectory group.

historical financial impact is negligible compared to standard inland tornadic activity. For an insurance agency, these figures suggest that "Sharknado-style" events represent an extremely rare risk rather than a primary cause of annual losses from catastrophes.

> **Note**
>
> A confidence interval could not be calculated for the average damage of "Moving Inland", high shark-lifting potential tornadoes due to the extreme scarcity of data ($n = 2$). In these instances, the point estimate of \$91.5K per storm serves as a historical record rather than a predictive mean.

3.5 Additional Analysis

To further analyze the historical trends of tornado weather intensity, the (Enhanced) Fujita scale classifications for every recorded tornado from 1950-2025 were plotted on the line graph shown in Figure 3.5 below. In the figure, there is a clear spike in F0 and F1 tornadoes after 1995. Higher intensity tornadoes remained relatively constant throughout the time period. Further explanation for this feature in the data can be found in Section 4.1.

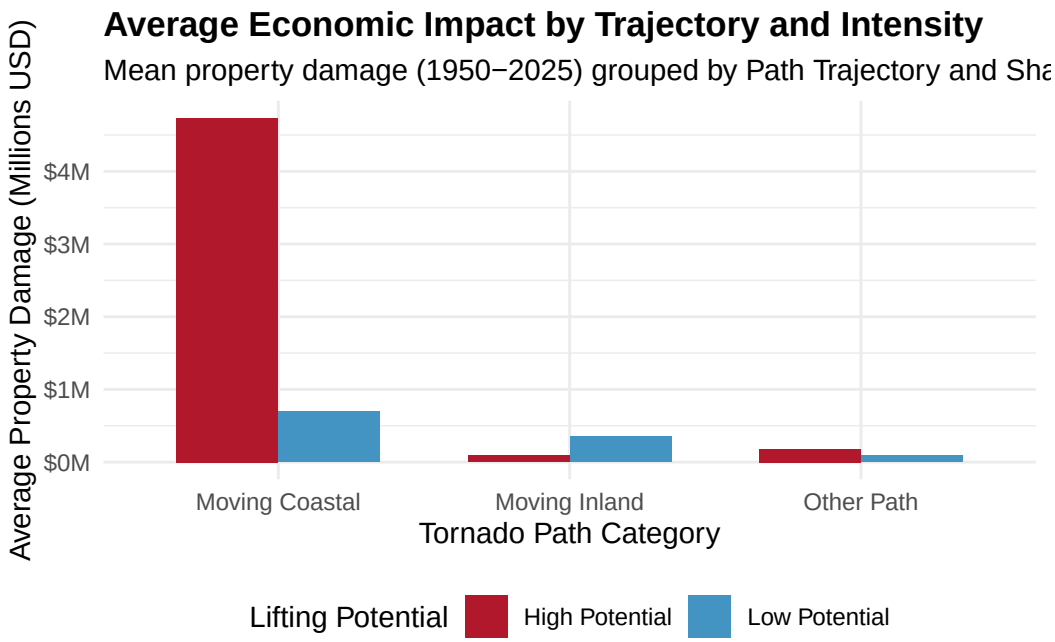

Figure 3.4: Categorized all tornadoes by trajectory and visualized the average property damage inflicted by high vs. low shark-lifting potential tornadoes within each path trajectory group.

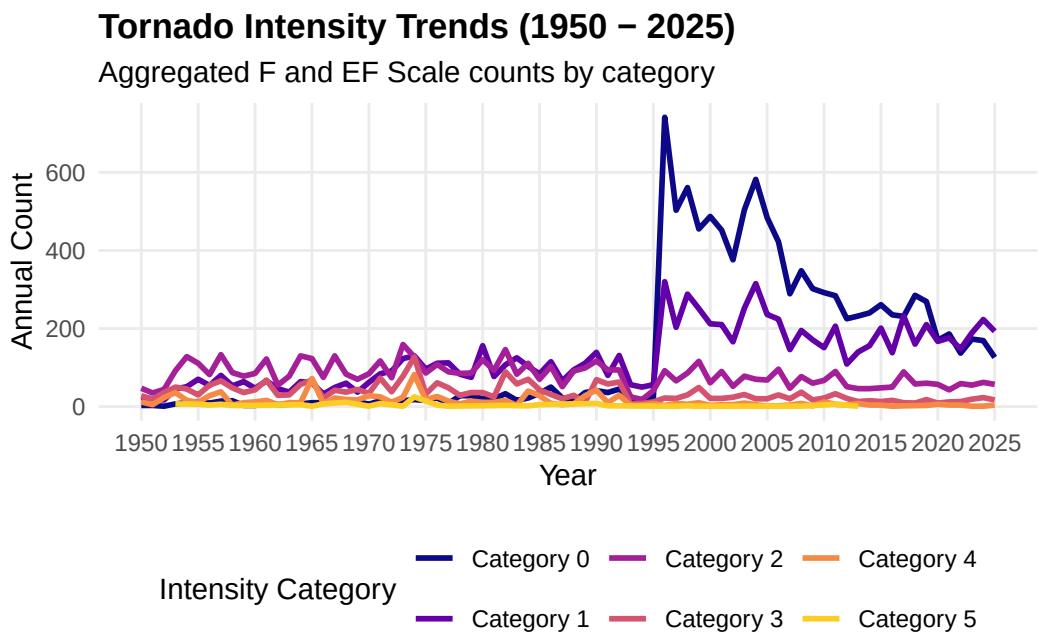

Figure 3.5: All tornadoes categorized by (Enhanced) Fujita category and counts are plotted from 1950-2025

4 Conclusion & Discussion

Using 75 years of tornado storm data from the NOAA, we analyzed the potential for sharknadoes to form in the US and the resulting economic impacts. We defined a storm as having the potential to produce a sharknado if it formed on the coast and moved inland ("Moving Inland") and had enough strength to carry a shark (category F3, F4, or F5). With this in mind, we sought to answer the research question:

What is the historical frequency, probability, and economic impact of high-intensity tornadoes (F3, F4, and F5) originating in coastal areas (Atlantic, Pacific, and Gulf of Mexico) and moving inland, compared to those following other trajectories?

4.1 Frequency & Probability of Sharknadoes

In the last 75 years, we found that only 2 out of 31,345 recorded tornadoes exhibited characteristics of "Moving Inland" and "High Potential" for shark lifting. Thus, the empirical probability of a sharknado-like storm is 0.0064% or approximately 1 in 15,673 U.S. tornado events (with a 95% confidence interval of 0.0017% to 0.023%). This frequency is significantly lower than that of high-intensity storms with a path classification of "Other Path", which occur at a rate of roughly 12.1%. Given that the concept of a sharknado originated from a fictional movie, this result was to be expected. For the insurance agency, the calculated statistical range confirms that while sharknado-like are rare outliers, the uncertainty in such a small sample size ($n = 2$) calls for treating sharknado risks as unpredictable long-tail volatility rather than a stable, predictable metric.

To further reduce concern regarding high shark-lifting potential tornadoes, we identified only 2 tornadoes with this intensity for both trajectory categories of "Moving Inland" (originating on the coast and moving inland) and "Moving Coastal" (originating inland and moving to the coast). This was expectedly insignificant compared to the 3,799 high shark-lifting potential tornadoes in the "Other Path" trajectory category, which makes sense since the inland Tornado Alley region breeds optimal conditions for high-intensity tornadoes compared to the coasts.

Additionally, the trends visualized in Figure 3.5 showed a dramatic spike in F0 and F1 tornadoes after 1995, and this is almost certainly due to the modernization of the National Weather Service, specifically the nationwide deployment of NEXRAD (Next-Generation Radar) Doppler technology from 1992 to 1997 (Administration 2025). Prior to this technology, fewer low-intensity tornadoes were recorded because they occurred in less populated areas or didn't

cause enough visible damage to be reported. It should also be noted that the frequency of high-intensity storms has remained relatively flat throughout all years of recorded data. Thus, there is no clear upward trend in storms capable of producing sharknadoes, even with increasing climate risk.

4.2 Regions of Concern

However, as previously mentioned, there has been an upward trend of tornadoes in atypical regions such as the Southeastern United States ("Tornadoes and Climate Change. National Geographic Education" 2023), and perhaps unexpectedly, the majority of tornadoes categorized as "Moving Inland" originated in the Atlantic or Gulf coasts near the southeastern side of the United States. With the southeast region already exhibiting these storm characteristics and with climate change creating volatility in extreme weather events, an increase in storms with sharknado potential in the southeast is not completely out of the question.

4.3 Economic Impact

In terms of the economic impact of sharknadoes, tornadoes with a path "Moving Inland" accounted for only 0.75% of all recorded tornado property damage, and even less of that was contributed by high shark-lifting potential tornadoes moving inland. Over 75 years, tornadoes "Moving Inland" with a high shark-lifting potential resulted in a total cumulative damage of $183K and an average of $91K per storm, $14.9K less than the average damage caused by all recorded tornadoes. Additionally, this was significantly less damage than tornadoes classified as "Other Path" with a high shark-lifting potential, which resulted in a total cumulative damage of $682M and an average of $179K per storm. This suggests that tornadoes originating in more typical regions (such as Tornado Alley) have historically caused far more damage overall and also on a per-storm basis. Thus, an insurance company covering property damages should be less concerned with sharknadoes, and more concerned with typical inland tornadoes.

4.4 Limitations & Future Exploration

4.4.1 Physics Preventing Sharkadoes

As mentioned in Chapter 1, it is nearly impossible for a tornadic water spout to pickup a shark out of the water. This is because tornadic water spouts don't typically suck up water, and the behavior of tagged sharks has shown that they usually dive deep when storms are approaching. So, while this report analyzes the frequency, probability, and economic impact of sharknado-like storms, the actual probability of a shark being lifted by a tornado remains

close to zero regardless of storm intensity. This study intentionally overlooks these mechanical improbabilities to provide a worst-case scenario baseline for the insurance agency.

4.4.2 Dataset Structure

The Storm Events Database (DOC/NOAA/NESDIS/NCEI 2026) was the core data used in this report. One of the major drawbacks of the data's structure is that a single storm may be recorded as multiple different storm segments. A new storm segment is created when the tornado enters a new county or zone, resulting in multiple data rows corresponding to one storm. Post-1996, segments belonging to the same storm have been connected using a unique identifier found in the column named `EPISODE_ID`, making the process of connecting segments relatively straightforward.

However, storms occurring before 1996 do not have a value in the `EPISODE_ID` column. To connect storm segments in this case, we matched two segments' dates, time, and start/end location coordinates. Because this is less precise than simply using a unique identifier to connect storm segments, there could be some room for error in the way full storm data was aggregated for storms occurring before 1996.

Further analysis could be conducted to compare the pre and post 1996 storms to ensure the difference in segment connection methods doesn't have any unforeseen effects.

4.4.3 Data Validity

The NOAA states that some of the data appearing in the Storm Events Database may be gathered from sources outside the National Weather Service. "An effort is made to use the best available information, but because of time and resource constraints, information from these sources may be unverified by the NWS." Therefore, the NOAA cautions users that the NWS does not guarantee accuracy or validity of the information (National Weather Service n.d.).

4.4.4 Data Biases

Due to significant advancements in meteorological technology and communication systems over the last 75 years, tornado data and records have certainly improved. In the early decades of the NOAA Storm Database, tornado records relied heavily on visual confirmation and manual reporting, which often resulted in under-reporting or less precise measurements in less populated regions. This could have an impact on time-series trends because the apparent increase in tornado frequency over time may be related to improved detection capabilities, as opposed to increased climate risk.

4.4.5 Sample Size & Temporal Constraints

While the dataset of $N = 31345$ events provides a solid foundation for analyzing modern tornado trends, the 75-year study period represents a relatively narrow window in terms of climate history. This limited temporal scope may not fully capture long-term climate patterns. Furthermore, the extreme rarity of "Moving Inland" and "High Potential" for shark-lifting ($n = 2$) limits our ability to make causal inferences or generalizations about the economic impact to future populations.

4.4.6 Storm Intensity Ratings

In the analysis, storm intensity ratings using the Fujita (F) scale and the Enhanced Fujita (EF) scale were consolidated into a single intensity metric (i.e., an F5 tornado was treated the same as an EF5 tornado). While the Enhanced Fujita Scale was derived from the original Fujita scale, there could be some unforeseen consequences as the scales are technically different. Treating an F5 tornado as exactly equivalent to an EF5 tornado may introduce a degree of measurement error, especially when comparing 20th-century storms to those occurring after the 2007 implementation of the EF system (National Weather Service 2015). This inconsistency could potentially skew time-series trends, as the criteria for a "High Potential" storm may have shifted slightly between the two time periods. However, for the purposes of this report, treating the scales synonymously is the most practical method for initial analysis.

4.4.7 Definition of "Coast" and "Other Path"

In the methodology, a rather arbitrary buffer of 3 miles was used to geographically and geometrically define the coast. The `st_buffer()` function from the `sf` package equally centers the buffer over a line (like the coastline), resulting in a zone that extends 1.5 miles into the ocean and 1.5 miles onto the land. Since only 73 out of 31,345 storms were identified as "Moving Inland" (originating on the coast and traveling inland), further analysis could be conducted by testing different buffer values, especially larger ones, to see if the recorded number of "Moving Inland" storms is affected.

Additionally, storm trajectories classified as "Other Path" included storms that started and remained inland, along with storms that started on the coast and remained on the coast. However, with the 3-mile buffer zone, storms that started on the coast and remained on the coast could technically include storms that originated within the 1.5-mile extension into the ocean and ended on land within the 1.5-mile extension inland. Thus, decreasing the buffer zone's size or moving the buffer zone closer to the water could shift some "Other Path" classifications to "Moving Inland". If this change were made, it is possible the results would show an increase in sharknado frequency.

4.5 Going Forward

Ultimately, with such a small number of recorded storms exhibiting characteristics expected of a sharknado along with relatively small amounts of inflicted damage, the insurance agency should not be concerned with sharknadoes having an impact on their bottom line. Though with climate change having an impact on extreme weather events, it wouldn't hurt the agency to periodically track if there is a continuing trend in storms forming in the southeastern region of the US, which is closer to the Gulf and Atlantic Coasts.

Action Items

- **Treat Marine-Origin Events as Rare Events:** High-intensity "Moving Inland" tornadoes remain an extreme statistical rarity, with only two recorded occurrences in 75 years. These should be categorized as low-probability, "long-tail" risks rather than primary drivers of annual catastrophe loss.
- **Maintain Standard Inland Risk Models**: With 97.86% of historical property damage occurring in "Other Path" trajectories, the agency should continue prioritizing traditional inland tornado risk models over specialized "Sharknado" or marine-origin coverages.
- **Monitor Southeast Regional Shifts:** While national "Sharknado" risk is negligible, the Gulf and Atlantic coasts account for the vast majority of coastal-to-inland storm paths. The agency should periodically track storm trends in the Southeastern U.S., where climate change *may* be increasing the frequency of atypical, high-intensity coastal events.

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A Technical Methodology

A.1 Data Pipeline

> **Note**
>
> See Figure A.3 for a diagram of the full data pipeline.

A.1.1 US Coastline Data

To implement the above definition of "Coast" and "Inland" (see Section 2.1), the R package `tigris` was used to provide a shapefile of the US Coastline. Once the original coastline shapefile was imported, it was filtered to only include the Atlantic, Pacific, and Gulf of Mexico coasts as requested in the original scenario prompt. The units of the coastline data were converted from degrees to meters using the `st_transform` function from the `tigris` package prepare for a meter-based buffer. Using the `st_buffer` function from the `tigris` package, a 3-mile (4828 meter) buffer was equally centered over the coastline, resulting in a zone that extends 1.5 miles into the ocean and 1.5 miles onto the land. Any point within this buffer zone is considered the "Coast" for the purposes of this analysis. Any point further inland than the buffer zone is considered "Inland". Finally, any coastline details smaller than 100m were smoothed using the `st_simplify` function from the `tigris` package to increase computational efficiency without sacrificing regional accuracy. A visual representation of this data can be seen in Figure A.1.

A.1.2 NOAA Storm Event Database

A.1.2.1 Pre-cleaning & Filtering

To begin processing the NOAA Storm Events Database, we `filter` the data to only include events that are `EVENT_TYPE = Tornado`. Additionally, any entries that are missing location coordinates for the features `BEGIN_LAT`, `BEGIN_LON`, `END_LAT`, `END_LON` were removed since we wouldn't be able to calculate their path trajectories. Finally, any features **not** in the following list were removed since they were not needed for the analysis:

- `EPISODE_ID`

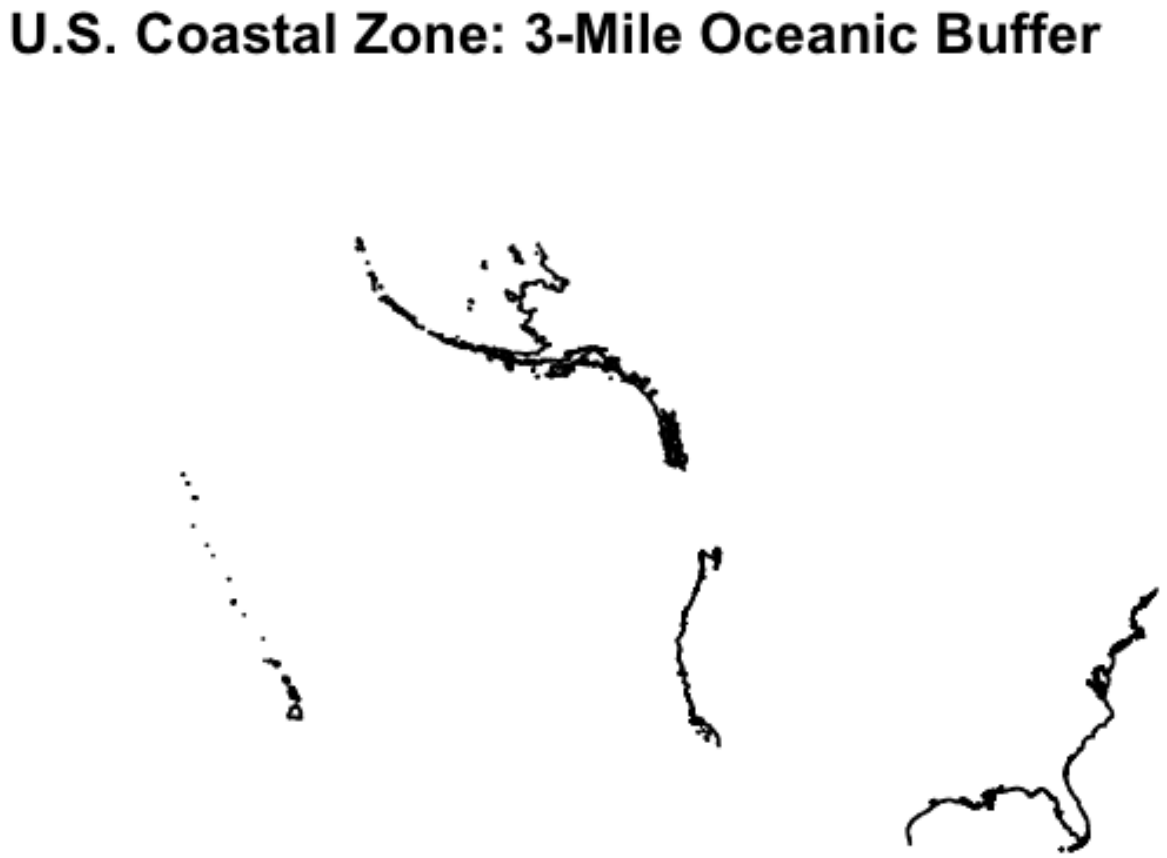

Figure A.1: This is a visual representation of the US Coastline 3-mile buffer zone generated in the code.

- EVENT_ID
- YEAR
- MONTH_NAME
- BEGIN_DATE_TIME
- STATE
- BEGIN_LAT
- BEGIN_LON
- END_LAT
- END_LON
- TOR_F_SCALE
- DAMAGE_PROPERTY

A.1.2.2 Geographical Classifications

Beginning the work needed to classify tornado path trajectories, we started by engineering two new features `Start_coast` and `End_coast`, which classify the tornado segment as starting/ending in the "Atlantic", "Pacific", "Gulf", or N/A. This is done by comparing the `BEGIN_LAT`, `BEGIN_LON`, `END_LAT`, `END_LON` features to the modified coastline shapefile.

From here, two more features were created `Start_location` and `End_location`, which classify the tornado segment as starting/ending on the "Coastal" or "Inland" based on whether the `Start_coast` and `End_coast` columns are empty for the feature.

A.1.2.3 Aggregate Storm Segments

NOAA Storm Events Database is structured so that a single storm may be recorded as multiple different storm segments. A new storm segment is created when the tornado enters a new county or zone, resulting in multiple data rows corresponding to one storm. Post-1996, segments belonging to the same storm have been connected using a unique identifier found in the column named `EPISODE_ID`, making the process of connecting segments relatively straightforward. However, storms occurring before 1996 do not have a value in the `EPISODE_ID` column. To connect storm segments from this era, we matched segments' dates, times, and start/end location coordinates.

Once tornado segments were stitched together into one full tornado entry, a new feature, `master_id`, was created to serve as an ultimate unique identifier for each full tornado. For pre-1996 storms, the unique `master_id` was formatted as `YEAR MONTH_NAME STATE`

BEGIN_LAT BEGIN_LON (e.g., 1950 September OKLAHOMA 35 -96.25). For post-1996 storms, the EPISODE_ID was used for the master_id since it is already a unique identifier for each full storm.

Because multiple tornado segments were stitched together, we needed to consolidate all of the data from each of the segments into a single entry. The following list summarizes how the data was consolidated for each feature:

- YEAR: use the year of the first segment
- MONTH_NAME: use the month name from the first segment
- Beginning_State: use the state of the first segment
- End_State: use the state of the last segment
- Start_location: use the start location of the first segment
- End_location: use the end location of the last segment
- Start_coast: use the start coast of the first segment
- End_coast: use the end coast of the last segment
- TOR_F_SCALE_MAX: maximum (Enhanced) Fujita scale ranking across all segments
- Total_Damage: sum of DAMAGE_PROPERTY across all segments

From here, a new feature path_movement was created, which classifies the path trajectory of the full tornado as "Moving Inland", "Moving Coastal", or "Other Path" based on the Start_location and End_location features. Below further describes the characteristics of each path trajectory, along with an illustration in Figure A.2.

- **Moving Inland**: tornado originated on the coast and moved inland.
- **Moving Coastal**: tornado originated inland and moved to the coast.
- **Other Path**: tornado originated either inland or on the coast and stayed in the same region of origin.

Finally, a new feature shark_lifting_potential was created, which classifies the shark-lifting potential of the full tornado as "High Potential" or "Low Potential", based on the TOR_F_SCALE_MAX feature. Below further describes the characteristics of these labels:

- **Low Potential**: tornado was given an F1 or F2 rating on the (Enhanced) Fujita Scale
- **High Potential**: tornado was given an F3, F4, or F5 rating on the (Enhanced) Fujita Scale

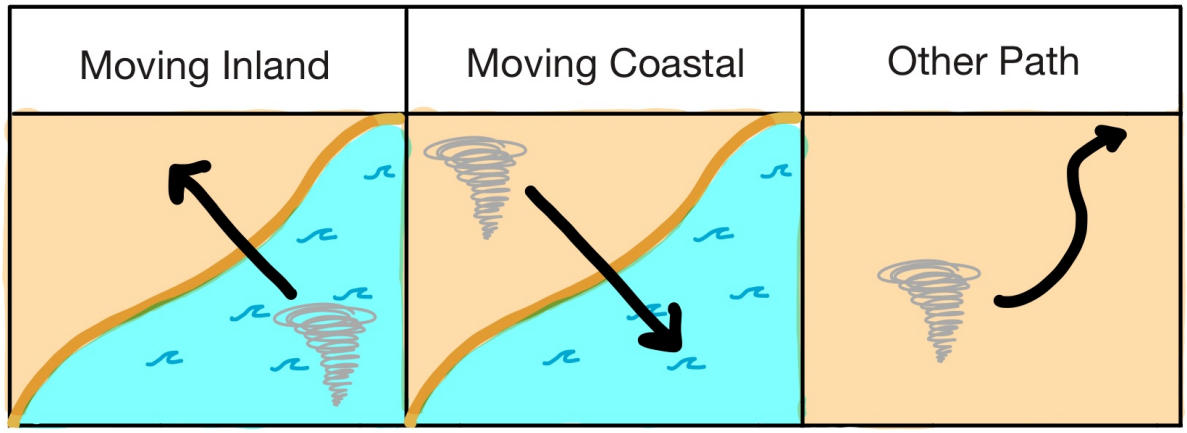

Figure A.2: This is a diagram illustrating the movement of tornadoes relative to the coast and inland, with path trajectory labels of “Moving Inland”, “Moving Coastal”, and “Other Path”.

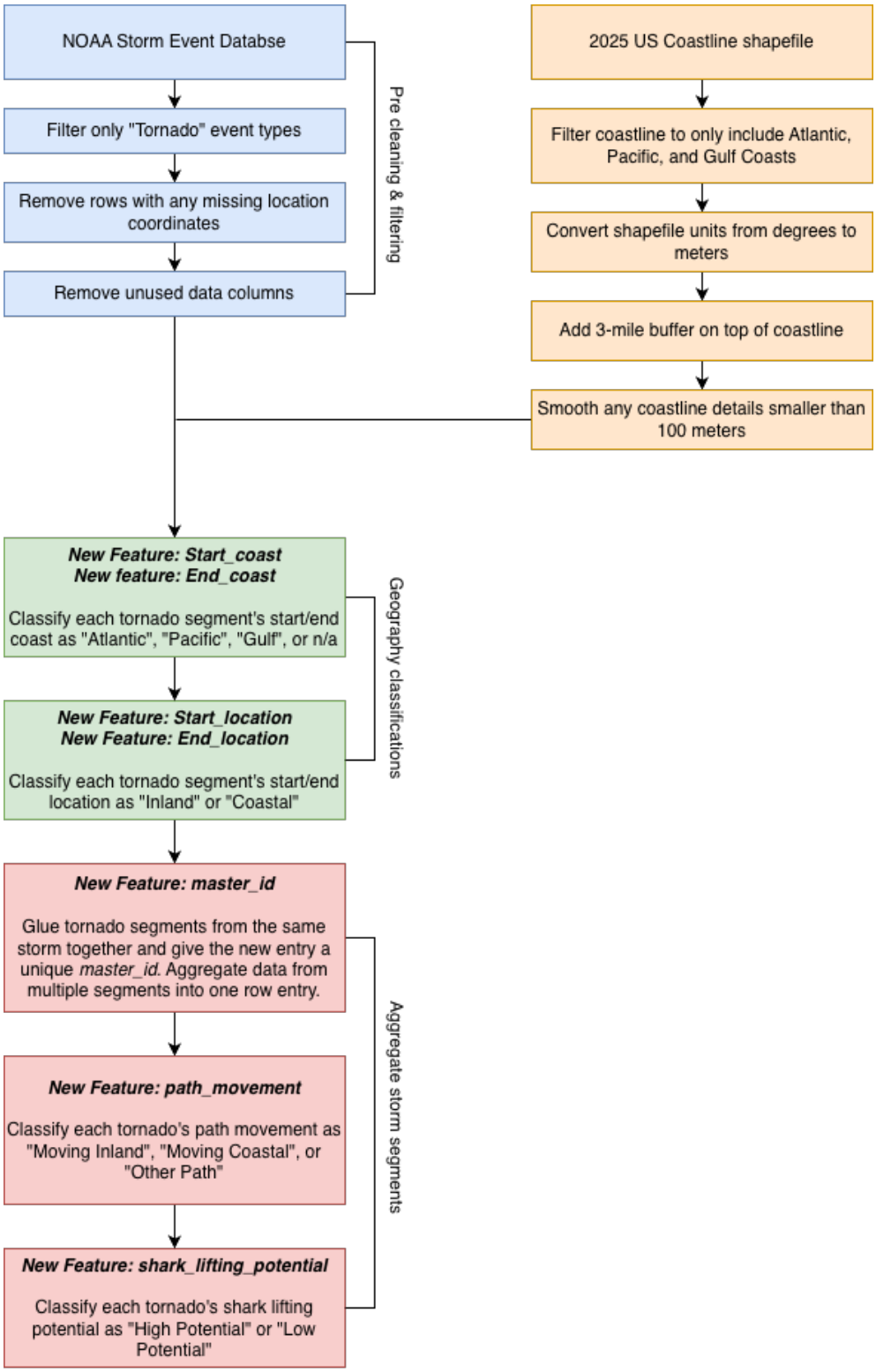

Figure A.3: This is a diagram illustrating the steps used to process data and engineer features from the NOAA Storm Events Database and US Coastline shapefile.

B Data Documentation

B.1 NOAA Storm Events Database

The Storm Events Database (DOC/NOAA/NESDIS/NCEI 2026) was entered by the National Centers for Environmental Information (NCEI) under the National Oceanic and Atmospheric Administration (NOAA), which receives data from the National Weather Service (NWS). This data set records the occurrence of intense storms and weather phenomena that have caused loss of life, injuries, significant property damage, and/or disruption to commerce. Also documented are rare, unusual weather phenomena that generate media attention, along with other record maximum or minimum meteorological events that occur in connection with another event. This data was collected and entered to create the official NOAA Storm Data publication and document significant weather events.

B.1.1 Collection

All data in the set was collected between January 1950 and October 2025. The National Weather Service receives its information from a variety of sources, which include but are not limited to: county, state, and federal emergency management officials, local law enforcement officials, skywarn spotters, NWS damage surveys, newspaper clipping services, the insurance industry, and the general public, among others. Because the NWS receives information from multiple sources, details are not provided on measurement instruments. New data for a given month is updated in the database 75-90 days after the end of that month.

B.1.2 Structure & Formatting

Since 2006, NWS has supplied NCEI with data in the form of CSV text files that are imported into NCEI's Storm Events Database. In the data set, each year of data has three different CSV files related to each, where each file contains events from that given year and is characterized by one of three data categories/file types: Event Details, Locations, and Fatalities. This analysis only used the "Event Details" file for each year of data, as it contained all the needed data columns for analysis. Notable columns that span across the three file types include start and end date/times, location (state and periodic latitude/longitude coordinates), event type (e.g., tornado), intensity, and fatalities/damage.

Dates are recorded as *YYYYMM* or *DD-MM-YY HH:MM:SS*. Location coordinates are formatted in decimal degrees. Tornado intensities are ranked using the Fujita (F) scale or the Enhanced Fujita (EF) scale, which are categorical variables (F0-F5 or EF0-EF5).

Another notable characteristic of the data's structure is that a single storm may be recorded as multiple different storm segments. A new storm segment is created when the tornado enters a new county or zone, resulting in multiple data rows corresponding to one storm. For pre-1996 storms, there was no connecting identifier across storm segments belonging to the same full storm. For post-1996 storms, the `EPISODE_ID` was used as the connecting identifier across storm segments belonging to the same full storm.

As an additional note, the Storm Events Database main page states that, "NCEI has performed data reformatting and standardization of event types but has not changed any data values for locations, fatalities, injuries, damage, narratives, and any other event-specific information."

Repository
README - information on the naming conventions of the files.
Storm Data Export Format - description of columns across Event Details Files, Location Files, and Fatality Files

B.1.3 Data Validation & Quality Control

According to the "Storm Data FAQ Page" (National Weather Service n.d.), "An effort is made to use the best available information, but because of time and resource constraints, information from these sources may be unverified by the NWS. Therefore, when using information from Storm Data, customers should be cautious as the NWS does not guarantee the accuracy or validity of the information."

B.1.4 Data Reuse

As a product of the U.S. Federal Government (Department of Commerce), this data is in the public domain and is free to use for research without a paid license.

B.2 U.S. Census Mapping Files

`tigris` is an R package that allows users to directly download and use TIGER/Line shapefiles from the US Census Bureau (Walker 2025). This analysis utilized `tigris` to retrieve a current shapefile of the 2025 United States coastline. In the analysis, this shapefile is used to establish the study's geographic boundaries between the coast and the inland.

The Census Mapping Files are entered by the U.S. Census Bureau. Broadly, these files are considered as Geographic Information System (GIS) mapping files, which contain spatial data

encoded into a file format. Examples of these types of files include Shapefiles, KML, and File Geodatabase files. For this report, the "Coastline National shapefile" held data points that mapped the coastline of the U.S. TIGER stands for Topologically Integrated Geographic Encoding and Referencing system and represents the U.S. Census Bureau's geographical spatial data.

The purpose of the U.S. Census TIGER/Line Shapefiles is for "statistical data collection and tabulation purposes only. Their depiction and designation for statistical purposes does not constitute a determination of jurisdictional authority or rights of ownership or entitlement and are not legal land descriptions," [Bureau (2025a)}. An additional disclaimer: "TIGER/Line Shapefiles do not include demographic data, but they do contain geographic entity codes (GEOIDs) that can be linked to the Census Bureau's demographic data," (Bureau 2025b).

More technical documentation regarding the U.S. Census Bureau TIGER/Line Shapefiles can be found at TIGER/Line Shapefiles and TIGER/Line Files Technical Documentation.

B.2.1 Collection

The Census Bureau maintains the MAF/TIGER System (Master Address File/Topologically Integrated Geographic Encoding and Referencing) through several sources, such as federal, state, and local governments, the Boundary and Annexation Survey (BAS), aerial imagery and fieldwork, and historical sources (Bureau 2025a).

B.2.2 Structure & Formatting

The Coastlines Shapefile is a national-level dataset provided as a zipped archive containing seven specific GIS file types, such as .shp for geometry and .dbf for tabular attributes. The data is formatted using the GCS NAD83 coordinate system with six decimal places of precision, ensuring high spatial accuracy for coastal mapping. Within the attribute table, each linear feature is standardized using the L4150 MAF/TIGER Feature Class Code and includes a primary name field corresponding to the bordering body of water (Bureau 2025a).

B.2.3 Data Validation & Quality Control

The Census Bureau maintains data quality through the Boundary and Annexation Survey (BAS), an annual program that verifies legal boundaries with tribal, state, and local governments to ensure the MAF/TIGER System remains current. Additionally, the Bureau employs automated validation edits and provides a Spatial Metadata Identifier (SMID) for every line segment, allowing users to track the specific source and horizontal accuracy of the data (Bureau 2025a).

B.2.4 Data Reuse

"Copyright protection is not available for any work of the United States Government (Title 17 U.S.C., Section 105). Thus, you are free to reproduce census materials as you see fit. We would ask, however, that you cite the Census Bureau as the source," (Bureau 2025a).